Butch's Quick Start Guide to String Building

This guide will get you started building your own compound strings. It provides tips and tricks as well as Butch's wisdom and experience.

MATERIALS

String

Most used:	BCY 452x
Less hand shock:	BCY X99
Soft on shots:	BCY Mercury 2
Waxless (not slick):	Bloodline Rev 99
Old wood limbs:	BCY B55
With steel cables:	BCY B55

Serving

Cam serving:	BCY Halo .014
End loops:	BCY Halo .007
Narrow cams:	BCY Powergrip .009
Center serving:	Angel Majesty .018
	Angel Majesty .021

TOOLS

Layout

- 3 post set up - used to serve one end then flip to do other end

OR

- 4 post - used for layout and serve both end loop without moving
- Layout spooler - makes the layout more consistent tension
- Flemish jig - makes Flemish twist strings easier, ultimate length control
- Spool holder - makes easy to layout strings
- Post setting spreadsheet - gets twist and post settings after shrinkage of twists
- String separators – separates colors while twisting

Serve

- Serving bobbin/jig (AAE, Beiter, Bear Paw)
- String clamps - helps serve tighter on long strings (mostly center servings)
- Serving machine - Specialty or Bloodline

Stretch

- Stretcher/Twister- used for making and measuring strings
- Air stretcher - used with air compressor usually 100 psi system
- Manual stretchers - used for more quiet operation

Misc

- Lighter - burning tags
- Razor blades or knife
- Pliers - for pulling tags

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LAYOUT PROCESS

1. Determine string length needed based on twists and creep - see spreadsheet
2. Set 2, 3 or 4 post to measurement
3. Lay out desired number of strands wrapping with a consistent tension - number of strands based off manufacture specs.
4. Serve end loops, either tag end serving or serving like .007 halo

STRETCH/TWIST PROCESS

1. I like to stretch to 300 lbs prior to twisting for about 10 minutes
2. Twist string to .67 ratio per inch (can vary per builder). Install string separators. Do not let the tension exceed 200 lbs on the twister scale. I will loosen the knobs and use back pressure with my hand while twisting to 80% of the total and then remove string separators and finish string twists. Take fingers and rub the string to even out final twists making them even.
3. After twisting I will install on stretcher and crank up to 400 lbs and leave set for 10 minutes. After 10 minutes or so reduce tension to 50-100 lbs. I will crank it back up to 400 and leave for a few hours while monitoring the decrease in poundage on the scale. It shouldn't drop much after a coupe hours, but could vary greatly per builder. (15 minutes to 12 hours)
4. After stretching twisted string, remove tension and allow string to relax for 10 minutes.
5. Measure the string at 100 lbs and adjust length to match desired string length. Note that serving can and will add length to string up to .125" depending on wax and length of string so take that into account.

SERVING PROCESS

1. Serving in the proper direction is very important. Always serve the direction that will add twist to string. Take you hand and apply a twist in the direction the serving jig will be forcing the string to add twists.
2. Set tension to 350 lbs on the stretcher/twister.
3. Set the serving bobbin/jig tension. Tighten to the point the serving tension will almost add twists to the string. Not that too much tension will break or cause peep rotation.
4. Serve the string from one end to the other in that order - loop towards center, string stop, center serving, cam serving from center to loop.
5. Install on bow

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WHERE TO GET

String materials & serving

- Lancaster Archery Supply
- 60x
- bowstringshop.com
- BCY fibers (must be a dealer)
- Bloodline

Speed nocks

- red and black from Saunders archery
- speed stacks from Gas bow strings (Eric Griggs) or bowstringshop.com

Custom heat shrink (for speed nocks)

- Daniel Digges (Facebook)
- bowstringshop.com
- Gas bow strings
- J W Pskowski (Facebook)

String serving specs

Archery talk forum has a bunch in the arrows and strings section. That is a take one leave one basis and I recommend participating to help others. Some bow manufactures will share on their websites. PSE and Hoyt will have specs.

String jigs

Of course Baker Archery Products! A few others manufacture jigs and home made jigs.

Serving machines

- Specialty Archery Super Server
- Bloodline Python Server
- Apple server (can sometimes be found used)
- HTM server (can sometimes be found used)